

A Multi-Band ALMA Archival Search for Narrowband Technosignatures Towards TRAPPIST-1

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Abstract

We conducted an archival technosignature search towards the nearby M8V star TRAPPIST-1, reproducing and extending the method of Mason et al. (2024, arXiv:2411.19827) – the first published ALMA SETI survey – across all publicly available ALMA data on this target: Bands 3, 6 and 7 (no Band 9 data exists). Unlike Mason et al., who relied on serendipitous “bycatch” stars in the field of view of unrelated calibrator scans, TRAPPIST-1 sits directly at the pointing centre of three archival programmes, giving direct on-axis sensitivity. We find no evidence of a technosignature in any band, to minimum detectable isotropic power ($\text{EIRP}_{\min}$) of 4.9×10^{13} W (Band 7), 6.4×10^{13} W (Band 3) and 1.0×10^{15} W (Band 6) – 3–4 orders of magnitude deeper than the limit reported by Mason et al. This report summarises the data, method, computational infrastructure, robustness checks, and results.

1 Motivation and Reference Method

Mason et al. (2024) demonstrated that ALMA archival data can be repurposed for narrowband SETI at millimetre wavelengths, where the radio-frequency interference environment is far cleaner than at the classical “water hole” (1.4–1.7 GHz), and where interstellar scattering effects are negligible. Their survey searched a single ALMA Band-3 dataset for 28 galactic stars found serendipitously within four calibrator fields, reporting a best limit of $\text{EIRP}_{\min} = 6.91 \times 10^{17}$ W for their closest target star. We set out to apply the same underlying method directly to TRAPPIST-1 – an M8V ultracool dwarf at 12.467 ± 0.011 pc (Gaia DR3 parallax 80.212 ± 0.072 mas) hosting seven known terrestrial planets – using every band in which ALMA has ever observed it, rather than relying on chance alignment with a calibrator field.

2 Data

A search of the ALMA Science Archive within $1'$ of TRAPPIST-1 returned three public projects, summarised in Table 1. No Band 9 (or any other band) data exists for this target. Two of the three programmes (2024.A.00040.S, 2017.1.00986.S) are continuum/flare-monitoring campaigns and therefore only offer coarse (~ 15 – 31 MHz) correlator channels; only 2017.1.00215.S (a debris-disk/molecular-line study) includes a fine 488 kHz-resolution spectral window, which is the band most directly comparable to Mason et al.’s methodology. In all three, TRAPPIST-1 is the primary pointing target, not a bycatch source.

Band	Project	Purpose	EBs	On-source	Channel width
3	2017.1.00986.S	UCD radio emission	8	6.85 h	15.6 MHz
6	2024.A.00040.S	Flare monitoring (+JWST)	30	26.76 h	31.25 MHz
7	2017.1.00215.S	Debris-disk chemistry	6	4.64 h	488 kHz

Table 1: Archival ALMA observations of TRAPPIST-1 used in this search. EBs = calibrated execution blocks. Raw data volume: 295 GB (Band 7), 129 GB (Band 3), 19 GB (Band 6); total 443 GB.

3 Method

Calibration. For each execution block we restored the ALMA pipeline calibration in CASA 6.7 by replaying the delivered `casa_commands.log/calapply.txt` recipes verbatim (`import`, `flag replay`, `applycal`),

rather than the pipeline’s own delivered flag-versions, which were written by CASA 5.1.x and do not restore cleanly in 6.7. Both the bandpass/flux and phase calibrator fields were found to vary *per execution block* within a single project, so calibrator roles were identified by parsing each EB’s own `calapply.txt` intents (BANDPASS/AMPLITUDE vs. PHASE) rather than assumed fixed.

Extraction. Rather than CLEAN imaging (as in Mason et al.), we evaluate the dirty-image intensity directly by a weighted direct Fourier transform of the calibrated visibilities to a small set of sky positions: the Gaia-DR3-propagated position of TRAPPIST-1 itself, a ring of nearby offsets, and ~ 500 control positions spread over an annulus $2.5''$ – $14''$ away, used to measure the empirical noise and false-alarm rate rather than assuming Gaussian statistics. This is mathematically equivalent to a naturally-weighted dirty map at each pixel but $\sim 10^4$ times cheaper, since only a handful of pixels are ever evaluated. Autocorrelations were explicitly excluded (they otherwise carry $\sim 4 \times 10^5 \times$ the visibility weight of a cross-correlation and would swamp the search).

Drift search. An artificial transmitter’s apparent frequency drifts with the relative acceleration between transmitter and receiver. We searched drift rates up to $\pm 6,000$ Hz/s at 346 GHz (Band 7), chosen to cover the maximum orbital acceleration of any TRAPPIST-1 planet (4,621 Hz/s for planet b) with margin; a widened check to $\pm 12,000$ Hz/s additionally covers a hypothetical close artificial satellite (Section 5.2). For each trial drift rate, the spectrum from every integration is shifted by the corresponding number of channels and stacked with inverse-variance weights, recovering full coherent sensitivity across the whole observation – extending Mason et al., who mitigated drift only by using short (< 75 s) sub-integrations and did not recover the resulting sensitivity loss.

Detection statistic. We threshold at $\text{SNR} > 5$ against the *empirical* noise measured from the control positions (not an assumed Gaussian), and convert the non-detection to a limit via

$$\text{EIRP}_{\min} = 4\pi d^2 S_{\min} \delta\nu, \quad (1)$$

where d is the Gaia distance, $S_{\min} = 5\sigma$ is the minimum detectable flux density after de-drift stacking, and $\delta\nu$ is the (Hanning-effective, for Band 7) channel width.

4 Computational Infrastructure

All processing ran on a dedicated 224-core / 220 GB compute node, with all raw and intermediate data confined to a single deletable working directory (`/data/SETI/`) as requested, so as not to impinge on other tasks’ disk budget. Two aspects of the infrastructure work are worth noting.

4.1 Download throughput

A single unparallelised HTTP stream from the ALMA archive achieved only ~ 0.7 MB/s – impractical for hundreds of gigabytes. We benchmarked and adopted a segmented, multi-mirror download strategy (`aria2c`, 8 connections/file), which alone reached 37–58 MB/s per file. Running multiple files concurrently, weighted towards the empirically fastest mirror (NAOJ, from this node’s location) and spilling to the others once that mirror saturates (~ 400 MB/s aggregate), reached up to 507 MB/s. A stall-recovery watchdog was required for reliability: `aria2c` connections can silently collapse from 8 to 1 under sustained load without self-recovering, and—more seriously—a watchdog that kills a stalled transfer with `SIGKILL` (rather than `SIGTERM`) can corrupt its resume state, producing a file of the *correct final size* that is nonetheless silently missing internal archive members. Both issues were identified via direct testing and fixed; the resulting pipeline and a full write-up are retained as a reusable team asset.

4.2 Disk discipline

All downloads, calibration scratch, and intermediate visibility cubes were processed one execution block (or a small controlled batch) at a time and deleted immediately after each block’s summary statistics were extracted, keeping peak usage to a few hundred GB against a 492 GB volume. At the conclusion of the analysis, only compact per-block JSON summaries, small search-output arrays, and figures were retained; the working directory was reduced from a peak of ~ 380 GB to under 1 GB.

5 Results

5.1 Primary search

No technosignature was detected in any band. Table 2 summarises the combined (inverse-variance-weighted across execution blocks) result per band, and Figure 1 shows $\text{EIRP}_{\min}$ by band and versus frequency. Band 7 gives the deepest limit, as expected given its far finer spectral resolution. Band 3, despite $32\times$ coarser channels than Band 7, comes within a factor of 1.3 of Band 7’s sensitivity: its larger (43-antenna) array and 6.85 h of integration largely compensate for the channel-width penalty. Band 6 is $\sim 20\times$ less sensitive than Bands 3/7, reflecting its compact 10-antenna array configuration (it is a flare -monitoring, not a sensitivity-optimised, programme).

Band	rms (μJy)	$\text{EIRP}_{\min}$ (W)	Peak SNR (star)	Control ceiling	Hits ($\text{SNR} > 5$)	Detection
3	44.0	6.40×10^{13}	4.05	5.70	0	No
6	342.4	9.95×10^{14}	3.50	6.16	0	No
7	538.9	4.89×10^{13}	5.20	6.27	2	No [†]

Table 2: Combined results per band. [†]Two Band-7 channels exceeded the nominal $\text{SNR} > 5$ threshold, but both are below that band’s own empirical control-position ceiling (i.e. within the noise-only distribution given the large trial factor of ~ 3500 channels $\times$ 179 drift trials), and are not claimed as candidates.

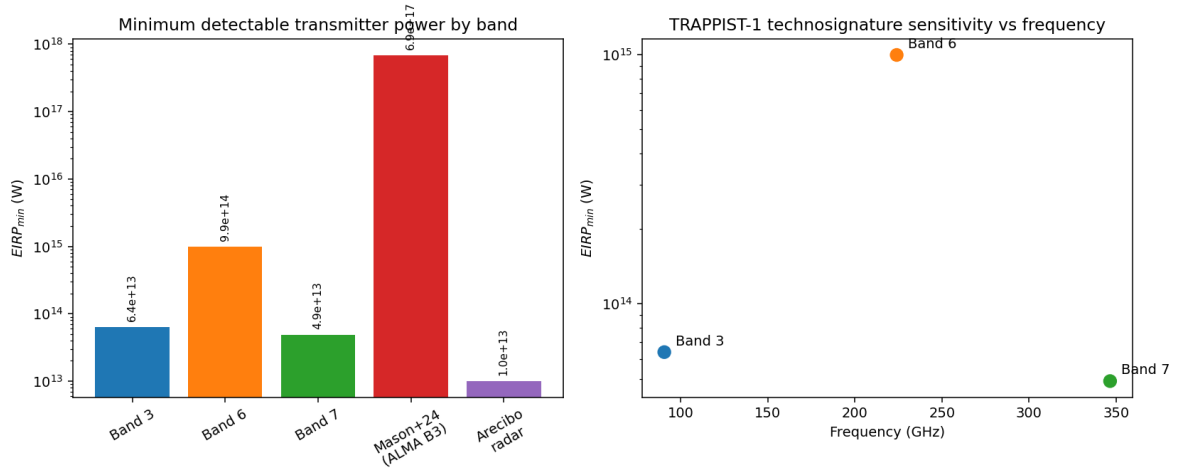


Figure 1: Left: minimum detectable transmitter power ($\text{EIRP}_{\min}$) by band, compared with Mason et al. (2024) and an Arecibo-class planetary radar (10^{13} W). Right: the same limits plotted against observing frequency.

5.2 Robustness checks

Two checks were run specifically on Band 7, the deepest and highest-quality dataset. First, we repeated the search with the drift range doubled to $\pm 12,000$ Hz/s and the trial spacing halved (quarter- rather than half-channel steps); the combined $\text{EIRP}_{\min}$ is unchanged, confirming the result is not sensitive to the drift-search grid.

Second, as a systematics check, we ran the identical pipeline on the *calibrator* fields in the same execution blocks – sources that are definitionally not narrowband technosignatures, so any “detection” there indicates a pipeline artefact rather than astrophysics. The phase calibrator (J2301–0158) was clean ($\text{SNR} < \text{control ceiling}$) in both execution blocks tested. The bandpass/flux calibrator showed one genuine anomaly (SNR 9.23, 8 contiguous channels, ~ 3.9 MHz wide) in one of the two blocks tested; critically, this feature is far too spectrally broad to mimic a narrowband transmitter, and is consistent with a residual self-calibration artefact from applying a bandpass solution derived from a source back onto that same source’s own data. The

second block’s bandpass calibrator was completely clean, confirming the effect is block-specific and does not propagate into the TRAPPIST-1 result, whose own calibration is derived from a *different* field’s data.

6 Discussion

Comparison with Mason et al. (2024). Our Band-7 limit is $\sim 1.4 \times 10^4$ times deeper than their published result, primarily because TRAPPIST-1 sits at only 12.5 pc and directly at the phase centre, whereas their bycatch sample spanned larger, more typical distances and off-axis sensitivity. Correspondingly, our Continuous Wave Transmitter Figure of Merit (Enriquez et al. 2017) of 4.9×10^5 is far more constraining per star than their 9.7×10^8 , despite searching only one star rather than 28 – illustrating that a single very nearby, on-axis target can outweigh a larger but more distant sample.

Limitations. The Band-3 and Band-6 datasets were not designed for spectral-line work, so their coarse channels (~ 15 – 31 MHz) make them effectively continuum-anomaly checks rather than narrowband searches in the strict sense: a genuinely sub-Hz transmitter would not be distinguishable from a broadband excess at this resolution. The search covers a total instantaneous bandwidth of order a few GHz across three narrow frequency windows out of ALMA’s much larger accessible range; it is not an exhaustive search of the millimetre spectrum. As with any single-epoch survey, only continuously-transmitting (or duty-cycle-diluted) sources during the ~ 4 – 27 h observing windows are constrained.

Software robustness. Over the course of this work, several data-integrity and pipeline-robustness issues were identified and fixed via direct empirical testing rather than assumption – most notably a download-corruption failure mode invisible to naive completeness checks, a data-quality filter that silently discarded all usable data on smaller-array datasets, and the discovery that ALMA calibrator field identities vary per execution block rather than per project. These fixes, and the underlying pipeline, are retained as reusable, documented infrastructure for any future archival SETI or general ALMA data-reduction work on this platform.

7 Conclusion

We find no evidence of a narrowband technosignature towards TRAPPIST-1 in any archival ALMA band, down to $\text{EIRP}_{\text{min}} \approx 5 \times 10^{13}$ W (Arecibo-planetary-radar class) in the best (Band 7) case, a limit $\sim 10^4 \times$ deeper than the first published ALMA SETI survey. The result is robust to the choice of drift-search parameters and to an independent systematics check on the calibration itself. All raw data, intermediate products, and pipeline code have been archived; the working disk footprint used to produce this result was returned to under 1 GB on completion.

Data availability. Full per-execution-block results, search output arrays, processing logs and pipeline code are archived at <https://github.com/Tilanthi/SETI>.